Given:

1. $b = \neg b_{1,1} \wedge b_{1,2} \wedge b_{2,1}$

Known $= \neg P_{1,1} \wedge \neg P_{1,2} \wedge \neg P_{2,1}$ $\Rightarrow$ Explored squares [1,1], [1,2], [2,1]

Each squares independently includes a pit, which is 0.01 & $q = 0.99$

According to breeze Rule, $b_{1,2} = \text{true} \Rightarrow P_{1,3} \vee P_{2,2}$ // $b_{2,1} = \text{true} \Rightarrow P_{2,2} \vee P_{3,1}$

Let $x = P_{1,3}$ // $y = P_{2,2}$ // $z = P_{3,1}$

Let E as all evidence $\Rightarrow$ (breezes + known)

$\Rightarrow E = (x \vee y) \wedge (y \vee z)$

$x(1,3)$	$y(2,2)$	$z(3,1)$	Prior P	$x \vee y$ & $y \vee z$
0	0	0	×	×
0	0	1	×	×
0	1	0	pq^2	0
0	1	1	p^2q	0
1	0	0	×	×
1	0	1	p^2q	0
1	1	0	p^2q	0
1	1	1	p^3	0

$Z = pq^2 + 3p^2q + p^3 = 0.010099$

$P(P_{1,3} \mid known, b)$

$$P(x=1 \mid E) = \frac{2p^2q + p^3}{Z} = \frac{2 \times 0.01^2 \times 0.99 + 0.01^3}{0.010099} \approx \underbrace{0.0197}_{p}, \underbrace{0.9803}_{q}$$

$P(P_{2,2} \mid known, b)$

$P(P_{2,2} \mid known, b) = 0.01$

$$P(y=1 \mid E) = \frac{0.01}{0.010099} \approx \underbrace{0.9902}_{\text{Pit}(p)}, \underbrace{0.0098}_{\text{No Pit}(q)}$$

For Logical agent

$\Rightarrow (P_{1,3} \vee P_{2,2}) \wedge (P_{2,2} \vee P_{3,1})$ // It doesn't know which one is more dangerous and needs to pick one from [1,3] and [2,2] randomly. Thus, it's about 50% chance of death and survive.

For probabilistic agent

$\Rightarrow P(1,3) \approx 2\%$ and $P(2,2) \approx 99\%$ of death. Thus, it will move to [1,3] since it's about 2% of chance of death and 98% chance of survive.

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(a) it's (C). Since All 3 G-nodes are mutually independent and has no Parents

(b) Network (a) and (b) make independence claims that are consistent with the hypothesis about the inheritance handness because G_{child} depends on the G_{father} and G_{mother} that matches the inheritance hypothesis.

(c) (a) is the best description of the hypothesis. (b) can imply Same thing like (a), but it allows extra dependencies and influences of parents' handness on child's handness.

(d)

G_{father}	G_{mother}	l	r
l	l	$1-m$	m
l	r	0.5	0.5
r	l	0.5	0.5
r	r	m	$1-m$

(e) $\sum_{g_m, g_f} P(G_{child} = l \mid g_m, g_f) P(g_m, g_f)$

$= \sum_{g_m, g_f} P(G_{child} = l \mid g_m, g_f) P(g_m) P(g_f)$

$= (1-m)q^2 + 0.5q(1-q) + 0.5(1-q)q + m(1-q)^2$

$= q^2 - mq^2 + q - q^2 + m - 2mq + mq^2$

$= q + m - 2mq$

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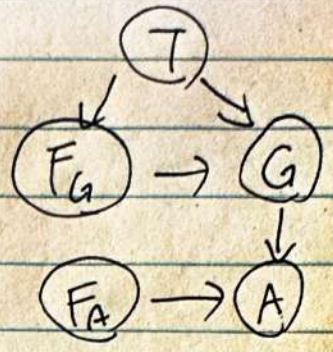
(a) A: alarm sounds

F_A: alarm is faulty

F_G: gauge is faulty

G: gauge reading

T: actual core temperature



(c)

T	F_G	G = High	G = Normal
High	False	x	$1-x$
Normal	False	$1-x$	x
High	True	y	$1-y$
Normal	True	$1-y$	y